

Document 9 – Data Dictionary

1. Overview

This document provides the data dictionary for the Smart Destination Recommender dataset. The purpose of this document is to define the structure, meaning, and expected values of each dataset used in the MVP.

The data dictionary supports data governance, transparency, maintainability, and future scalability of the solution.

The dataset consists of five core entities:

- destinations
 - sustainability_scores
 - congestion_scores
 - users
 - bookings_history
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2. destinations.csv

Description

Master catalog containing tourist destinations available for recommendation.

Column Name	Data Type	Description
destination_id	String	Unique identifier of the destination
destination_name	String	Destination name
region	String	Spanish region or autonomous community
destination_type	String	Destination category (Beach, City, Nature, Mixed)
beach_score	Integer	Attractiveness for beach tourism (1–10)
culture_score	Integer	Attractiveness for cultural tourism (1–10)

nature_score	Integer	Attractiveness for nature tourism (1–10)
nightlife_score	Integer	Attractiveness for nightlife activities (1–10)
family_friendly_score	Integer	Suitability for family travel (1–10)
avg_price_per_day	Integer	Estimated average daily travel cost in EUR

Example

destination_id	destination_name
D001	Mallorca
D013	Barcelona

3. sustainability_scores.csv

Description

Contains sustainability indicators associated with each destination.

These indicators are used by the recommendation engine to promote sustainable tourism alternatives.

Column Name	Data Type	Description
destination_id	String	Destination identifier
carbon_score	Integer	Environmental impact score (0–100)
local_business_score	Integer	Support for local economy score (0–100)
public_transport_score	Integer	Public transportation quality score (0–100)
sustainability_score	Decimal	Overall sustainability score (0–100)

Business Usage

The recommendation engine may prioritize destinations with higher sustainability scores when sustainability is an important traveler preference.

4. congestion_scores.csv

Description

Contains monthly destination congestion estimates.

This dataset supports overtourism mitigation and destination balancing recommendations.

Column Name	Data Type	Description
destination_id	String	Destination identifier
month	Integer	Month number (1–12)
congestion_score	Integer	Estimated congestion level (0–100)

Interpretation

Score Range	Meaning
0–30	Low congestion
31–60	Moderate congestion
61–80	High congestion
81–100	Very high congestion

Example

destination_id	month	congestion_score
D001	8	100
D019	8	90

5. users.csv

Description

Contains synthetic traveler profiles representing major TUI source markets.

No real customer information is included.

Column Name	Data Type	Description
user_id	String	Unique traveler identifier
country	String	Country of origin
age_group	String	Traveler age range
budget_level	String	Budget category (Low, Medium, High)
travel_style	String	Preferred travel style
sustainability_preference	String	Interest in sustainable travel

Allowed Values

budget_level

- Low
- Medium
- High

travel_style

- Beach
- Culture
- Nature
- Family
- Nightlife
- Relax

sustainability_preference

- Low
- Medium

- High
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6. bookings_history.csv

Description

Historical booking dataset used to simulate traveler behavior and recommendation training data.

Column Name	Data Type	Description
booking_id	String	Unique booking identifier
user_id	String	Traveler identifier
destination_id	String	Destination identifier
booking_date	Date	Date when booking was made
travel_month	Integer	Planned travel month
stay_days	Integer	Length of stay
total_price	Decimal	Total booking value in EUR
user_rating	Integer	Customer satisfaction rating (1–5)

Example

booking_id	user_id	destination_id
B0001	U014	D001
B0002	U082	D004

7. Entity Relationships

The datasets are connected through the following relationships:

Users → Bookings

Destinations → Bookings

Destinations → Sustainability Scores

Destinations → Congestion Scores

These relationships allow the recommendation engine to combine traveler preferences, destination characteristics, sustainability metrics, and historical booking behavior.

8. Data Quality Rules

The following validation rules apply:

- All primary keys must be unique.
 - `destination_id` must exist in `destinations.csv` before being referenced elsewhere.
 - `user_id` must exist in `users.csv` before being referenced in `bookings_history.csv`.
 - Scores must remain within their defined ranges.
 - Null values are not permitted in key identifier fields.
 - Sustainability scores must be between 0 and 100.
 - Congestion scores must be between 0 and 100.
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9. Privacy and Compliance

All traveler records contained in the dataset are synthetically generated.

The dataset contains:

- No Personally Identifiable Information (PII)
- No real customer records
- No sensitive personal data
- No GDPR-restricted information

The dataset is intended exclusively for demonstration, prototyping, analytics, and AI recommendation experiments.